

# Scenario 1 - Level 1 (Bootstrapped / Ad Hoc) — Fully Self-Funded

## Scenario Summary

Maximum-constraint path. All capital comes from founders. Team operates at ~~40% of ideal utilization~~ (320 productive technical hrs/mo). Every phase stretches because work is heavily sequenced rather than parallelized, and operational workstreams compete with the same founder hours. No investor dilution, but runway risk is acute throughout.

- Economic cost to Production: **\$432,500 / INR 35,897,500**
- Target timeline to Production: **34 months**
- Founder capital required: **\$232,000 / INR 19,256,000**
- Investor capital: **\$0**
- Equity: 60 / 20 / 0 (20% investor pool reserved, unissued)
- Main risk to watch: founder burnout and cash exhaustion before revenue covers burn

## Timeline

| Phase                | Timing    | Duration  | Minor Checkpoints                                                                                                                                                                                                         | Gate to Advance                                                                                               |
|----------------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Pre-Pilot            | M0 - M8   | 8 months  | M1-M2: Architecture freeze, issue triage. M3-M4: APK backlog locked, doctor training outline. M5-M6: First internal rehearsals, callback SOP draft. M7-M8: Full internal test cycles, admin runbooks versioned.           | Core web flows stable; internal rehearsals pass; doctor and admin runbooks versioned; APK build underway.     |
| Pilot 1              | M8 - M12  | 4 months  | M9: Warm-network onboarding begins. M10: First paid consults (subsidized). M11: Callback calibration, doctor playbook v1 live. M12: Throughput measured, escalation paths tested.                                         | Ethiopia pilot runs live for 2-3 months with measured throughput, callbacks, and escalation handling.         |
| Pilot 2              | M12 - M22 | 10 months | M13-M14: Marketing push, user growth. M15-M16: Kenya prep, partner reporting. M17-M18: Roster planning, partial automation. M19-M20: Staffing calibration, refresher training. M21-M22: APK hardening, reporting cleanup. | 5k-10k user-capable operations with stronger APK, cleaner reporting, and staffed callback coverage.           |
| Production Readiness | M22 - M34 | 12 months | M23-M25: Compliance closeout, security hardening. M26-M28: Release freeze, launch playbooks. M29-M31: Certification refreshers, production support schedule. M32-M34: Final QA cycles, launch readiness review.           | Current prototype scope is production-safe across web and patient mobile with documented fallback operations. |

## Feature and Output Checklist

| Phase                | Output Checklist                                                                                                                                                                             |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-Pilot            | Web core hardened (sequentially); APK build started; doctor/admin training materials drafted; support desk scripts and callback trees defined; partner onboarding kits outlined.             |
| Pilot 1              | Patient app covers core journeys (limited scope); provider web is stable for GP flow; doctor roster active; admin callbacks manual but functional; manual fulfilment visible in-system.      |
| Pilot 2              | Core patient features complete; partner dashboards functional; staffing ladder active; reporting, payments, and support loops tightened; Kenya market research complete.                     |
| Production Readiness | Current prototype scope is production-grade across web and patient mobile; trained ops staff with manual fallback available; compliance documentation complete; release governance in place. |

## Services and SLAs

| Phase      | Uptime Target | Response Commitment            | Operating Model                                                                                                      |
|------------|---------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Pre-Pilot  | 95.0%         | Best-effort, same-day          | 3 core team + 7 part-time developers + 2 advisors; no employed clinical coverage; training and rehearsals only.      |
| Pilot 1    | 96.0%         | Same business day for critical | Minimum employed coverage: 2 doctors, 2 admin ops, 2 technical support. Partner clinics supplement.                  |
| Pilot 2    | 97.5%         | 4-hour P1 response             | Minimum employed coverage: 4 doctors, 4 admin ops, 2 technical support. Callback and partner reporting loops active. |
| Production | 98.5%         | 2-hour P1 response             | Minimum employed coverage: 6 doctors, 6 admin ops, 3 technical support for near-24/7 coverage.                       |

## Cost Breakdown

### Summary

| Category                                       | Amount                            |
|------------------------------------------------|-----------------------------------|
| Capital expenditure through Production         | \$218,400 / INR 18,127,200        |
| Operating expenditure before Production launch | \$214,100 / INR 17,770,300        |
| Total economic spend to Production             | <b>\$432,500 / INR 35,897,500</b> |

### Workstream Detail

| Workstream                              | Economic Cost (USD) | Economic Cost (INR) | Notes                                                                                                                                 |
|-----------------------------------------|---------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Product and founder office              | \$175,560           | INR 14,571,480      | 3 core team at \$20/hr x 88 hrs/mo x 34 months. Stretched timeline means more founder-month cost.                                     |
| Web platform hardening                  | \$14,900            | INR 1,236,700       | 745 hrs at \$20/hr senior rate. Same scope, just spread over more months.                                                             |
| Patient APK build                       | \$7,200             | INR 597,600         | Fixed quote: 480 hrs at \$15/hr blended.                                                                                              |
| QA, DevOps, security                    | \$6,600             | INR 547,800         | ~330 hrs at \$20/hr. Reduced from ideal due to budget constraint; higher residual risk.                                               |
| Admin and care operations               | \$148,240           | INR 12,303,920      | Ops staff ladder runs longer (34 months vs 20). Pilot 1 starts M8; ops burn accumulates over 26 months.                               |
| Partner onboarding and provider success | \$11,250            | INR 933,750         | 750 hrs parallel workstreams at \$15/hr blended. Timeline gives more calendar time but same effort.                                   |
| Legal, accounting, compliance           | \$8,500             | INR 705,500         | Longer timeline = more compliance touch-points. Friendly-network pricing at economic value.                                           |
| Infrastructure, phones, and tools       | \$38,750            | INR 3,216,250       | Monthly infra costs accumulate over 34 months. Pre-Pilot: ~\$100/mo; Pilot 1: ~\$400/mo; Pilot 2: ~\$700/mo; Production: ~\$1,150/mo. |
| Marketing, travel, rehearsals           | \$21,500            | INR 1,784,500       | Spread thinly across longer timeline. More travel cycles needed for partner retention over extended pilot.                            |

### Effort and Team

| Function               | Staffing Pattern                                                           | Notes                                                                                                    |
|------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Core team              | 3 part-time members @ <del>4 hrs/day each</del> (264 hrs/mo combined)      | Single owner/founder + 2 founding members. Time split heavily across product, ops, and fundraising prep. |
| Technical delivery     | 7 part-time developers, modeled at <b>~320 productive technical hrs/mo</b> | Lean utilization. Heavy sequencing — Android and web rarely run in parallel at full capacity.            |
| Advisory board         | 2 fractional advisors (~20-30 hrs/mo combined)                             | Finance, compliance, and strategic review.                                                               |
| Employed care coverage | Pre-Pilot: 0 / Pilot 1: 2 doctors / Pilot 2: 4 / Production Readiness: 6   | INR 60,000/mo per doctor. Partner clinics supplement but minimum employed coverage is explicit.          |
| Employed admin ops     | Pre-Pilot: 0 / Pilot 1: 2 admins / Pilot 2: 4 / Production Readiness: 6    | INR 18,000/mo per admin. Callbacks, exception handling, manual fulfilment.                               |
| Technical support      | Pre-Pilot: 0 / Pilot 1: 2 / Pilot 2: 2 / Production Readiness: 3           | INR 25,000/mo per tech support staff.                                                                    |

### Revenue and Pricing

#### Revenue Sources by Phase

| Revenue Source           | Pilot 1 (M8-M12)                               | Pilot 2 (M12-M22)                        | Production Readiness (M22-M34)   |
|--------------------------|------------------------------------------------|------------------------------------------|----------------------------------|
| GP consults              | Subsidized, Ethiopia-first, ~\$2-3/interaction | Full Ethiopia + Kenya pricing tests      | Core revenue line                |
| Specialist consults      | Minimal                                        | Grows after referral throughput improves | Important margin contributor     |
| Pharmacy commission      | Small, admin-assisted                          | Repeatable, still manual-heavy           | Stronger with partner conversion |
| Diagnostics commission   | Small, admin-assisted                          | Better once order flow consistent        | Retention lever                  |
| Travel/care coordination | Few cases, high touch                          | Higher-value but low-volume              | Strategic differentiator         |

#### Pricing Benchmarks

| Revenue Line             | Ethiopia Benchmark | Kenya Benchmark     | Notes                                                          |
|--------------------------|--------------------|---------------------|----------------------------------------------------------------|
| GP consult               | \$3.50 / ETB 200   | \$5.00 / KES 650    | Starts subsidized in Pilot 1; reaches benchmark by Production. |
| Specialist consult       | \$9.00 / ETB 513   | \$12.00 / KES 1,560 | Pilot 2 onward.                                                |
| Pharmacy take-rate       | 10% of order value | 10% of order value  | Pilot 1 onward with manual fulfilment.                         |
| Diagnostics take-rate    | 12% of order value | 12% of order value  | Pilot 1 onward with admin/lab coordination.                    |
| Travel/care coordination | \$10-15 per case   | \$10-15 per case    | Small volume, high-touch service.                              |
| Subscription (premium)   | Deferred           | Deferred            | Not modeled before meaningful scale.                           |

### Cash Flow and Runway

#### Monthly Burn by Phase

| Phase | Monthly Team Burn | Monthly Ops Staff | Monthly Infra | Monthly Total Burn | Revenue Offset |
|-------|-------------------|-------------------|---------------|--------------------|----------------|
|-------|-------------------|-------------------|---------------|--------------------|----------------|

|                                |         |         |         |          |                   |
|--------------------------------|---------|---------|---------|----------|-------------------|
| Pre-Pilot (M0-M8)              | \$5,930 | \$0     | \$100   | \$6,030  | \$0               |
| Pilot 1 (M8-M12)               | \$5,930 | \$2,482 | \$400   | \$8,812  | \$200-500/mo      |
| Pilot 2 (M12-M22)              | \$5,930 | \$4,361 | \$700   | \$10,991 | \$500-2,500/mo    |
| Production Readiness (M22-M34) | \$5,930 | \$6,542 | \$1,150 | \$13,622 | \$3,000-12,000/mo |

Monthly Cash Flow

| Month | Phase      | Founder Capital In | Revenue In | Outflow  | Net      | Ending Cash |
|-------|------------|--------------------|------------|----------|----------|-------------|
| M1    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$1,470     |
| M2    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$2,940     |
| M3    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$4,410     |
| M4    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$5,880     |
| M5    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$7,350     |
| M6    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$8,820     |
| M7    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$10,290    |
| M8    | Pre-Pilot  | \$7,500            | \$0        | \$6,030  | \$1,470  | \$11,760    |
| M9    | Pilot 1    | \$7,500            | \$200      | \$8,812  | -\$1,112 | \$10,648    |
| M10   | Pilot 1    | \$7,500            | \$350      | \$8,812  | -\$962   | \$9,686     |
| M11   | Pilot 1    | \$7,500            | \$400      | \$8,812  | -\$912   | \$8,774     |
| M12   | Pilot 1    | \$7,500            | \$500      | \$8,812  | -\$812   | \$7,962     |
| M13   | Pilot 2    | \$7,500            | \$600      | \$10,991 | -\$2,891 | \$5,071     |
| M14   | Pilot 2    | \$7,500            | \$800      | \$10,991 | -\$2,691 | \$2,380     |
| M15   | Pilot 2    | \$7,500            | \$1,000    | \$10,991 | -\$2,491 | -\$111      |
| M16   | Pilot 2    | \$7,500            | \$1,300    | \$10,991 | -\$2,191 | -\$2,302    |
| M17   | Pilot 2    | \$7,500            | \$1,600    | \$10,991 | -\$1,891 | -\$4,193    |
| M18   | Pilot 2    | \$7,500            | \$1,900    | \$10,991 | -\$1,591 | -\$5,784    |
| M19   | Pilot 2    | \$7,500            | \$2,100    | \$10,991 | -\$1,391 | -\$7,175    |
| M20   | Pilot 2    | \$7,500            | \$2,300    | \$10,991 | -\$1,191 | -\$8,366    |
| M21   | Pilot 2    | \$7,500            | \$2,400    | \$10,991 | -\$1,091 | -\$9,457    |
| M22   | Pilot 2    | \$7,500            | \$2,500    | \$10,991 | -\$991   | -\$10,448   |
| M23   | Production | \$5,000            | \$3,000    | \$13,622 | -\$5,622 | -\$16,070   |
| M24   | Production | \$5,000            | \$3,500    | \$13,622 | -\$5,122 | -\$21,192   |
| M25   | Production | \$5,000            | \$4,500    | \$13,622 | -\$4,122 | -\$25,314   |
| M26   | Production | \$5,000            | \$5,500    | \$13,622 | -\$3,122 | -\$28,436   |
| M27   | Production | \$5,000            | \$6,500    | \$13,622 | -\$2,122 | -\$30,558   |
| M28   | Production | \$5,000            | \$7,500    | \$13,622 | -\$1,122 | -\$31,680   |
| M29   | Production | \$5,000            | \$9,000    | \$13,622 | \$378    | -\$31,302   |
| M30   | Production | \$5,000            | \$10,000   | \$13,622 | \$1,378  | -\$29,924   |
| M31   | Production | \$5,000            | \$11,000   | \$13,622 | \$2,378  | -\$27,546   |
| M32   | Production | \$5,000            | \$12,000   | \$13,622 | \$3,378  | -\$24,168   |
| M33   | Production | \$5,000            | \$13,000   | \$13,622 | \$4,378  | -\$19,790   |
| M34   | Production | \$5,000            | \$14,000   | \$13,622 | \$5,378  | -\$14,412   |

Cumulative founder capital deployed: ~\$232,000 (\$7,500/mo x 28 months + \$5,000/mo x 12 months)

**WARNING:** Cash goes negative at M15. This path requires either (a) founders injecting additional lump-sum capital, (b) cutting ops staff earlier, or (c) accepting the path is not viable without some external funding. The negative cash position is the defining risk of S1-L1.

ROI and Break-Even

| Metric                                             | Value                                                                                                   |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Break-even month (monthly revenue >= monthly burn) | ~ <b>M29</b> (month 29 — revenue \$9,000 vs burn \$13,622; close but not yet; true break-even ~M32-M33) |

|                                                      |                                           |
|------------------------------------------------------|-------------------------------------------|
| Cumulative break-even (total revenue >= total spend) | Not reached within 34-month window        |
| ROI at 12 months post-production (M46)               | -35% to -25% (still recovering capital)   |
| ROI at 24 months post-production (M58)               | +5% to +20% (positive but modest)         |
| ROI at 36 months post-production (M70)               | +40% to +65% (healthy if growth sustains) |

Key Risks

| # | Risk                              | Likelihood | Impact   | Mitigation                                                                                                                                 |
|---|-----------------------------------|------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Cash exhaustion before break-even | Very High  | Critical | Founders must maintain reserve capital or accept negative cash position. Consider converting to S2/S3 if M15 cash trigger is hit.          |
| 2 | Founder burnout over 34 months    | High       | Critical | Part-time model helps, but sustained bootstrapping pressure without investor validation is demoralizing. Build in explicit rest cycles.    |
| 3 | Partner and clinic attrition      | High       | High     | Extended pilot duration risks losing early partners. Keep engagement through regular check-ins and incremental value delivery.             |
| 4 | Technology drift over 34 months   | Medium     | High     | Angular/Node ecosystem will shift significantly. Budget for at least one major framework upgrade cycle.                                    |
| 5 | Market window closure             | Medium     | High     | Competitors in Ethiopian and Kenyan health tech may establish during the extended timeline. Monitor and accelerate if threat materializes. |

Recommendation

**Not recommended as a standalone path.** S1-L1 is the survival-mode scenario. The 34-month timeline, negative cash position by M15, and accumulated founder burden make this the highest-risk combination in the matrix.

This path is viable only if:

- 1. Founders have personal savings exceeding \$250,000 and are willing to absorb negative cash flow for 12+ months
- 2. The team treats this as a side project with no urgency to capture market timing
- 3. There is a credible fallback plan to convert to S2 or S3 if the M15 cash trigger is hit

If none of these conditions hold, the founder should pursue S1-L2 at minimum or seek early-stage investment (S2 or S3). This scenario exists primarily as a lower bound for planning — it shows what happens if no outside capital materializes and the team operates at minimum capacity.

All figures reference params.md v2. Currency conversion: 1 USD = 83 INR.